

4(a)	(closed) double circulation ; capillary ; pulmonary vein ; right atrium ; A auricle septum ;	5
4(b)(i)	<i>two from:</i> 1 <i>idea</i> that (to be transported) many substances need to, dissolve / be in solution ; 2 ionic compounds / named, can, dissociate / dissolve ; 3 polar compounds / named, e.g. glucose / amino acids, can dissolve ; 4 globular proteins / named, e.g. antibodies, can dissolve ;	2
4(b)(ii)	<i>three from:</i> 1 water molecules attracted to each other ; A sticky / stickiness <i>cohesion:</i> 2 (hydrogen bonding provides) <u>cohesion</u> between water molecules ; A water is cohesive 3 reference to water leaving xylem (at top), pulling water (molecules below) ; A there is a transpiration pull <i>adhesion:</i> 4 <u>adhesion</u> to <u>cellulose</u> lining (of xylem) ; A <u>cellulose</u> wall 5 maintains / prevents falling of, column of water ; 6 AVP e.g. reference to cellulose hydrophilic / adhesion to hydrophilic parts of lignin ;	3